Count Me In TT: Building Speed, Accuracy & Confidence in Mathematics (2021–2025)

A proof-of-concept study on gamified math learning in Trinidad & Tobago

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Executive Summary

Between 2021 and 2025, *Count Me In TT* was tested by, Mathematics tutor, Mr Andre Burton, founder of A's Online Tutoring Services and creator of *Count Me In TT,* as an online session exercise and learning tool, designed to help students strengthen their multiplication fluency through repetition, recall, and timed practice. The goal was to measure not only improvement in speed and accuracy, but also the effect of gamified learning on students' motivation and confidence in mathematics.

A total of nine primary-level students, ranging from Standard Three to Standard Five, participated in the testing phase. Each student completed the digital multiplication exercise multiple times over several months, recording their completion times to track progress and personal growth.

The results were clear and encouraging. On average, students achieved a 50% improvement in completion speed after five attempts, with many reducing their times even further with continued practice. Importantly, every participant showed progress — both in measurable performance and in attitude toward mathematics.

Beyond the numbers, Mr Burton observed a noticeable shift in student behaviour. Learners became more enthusiastic, self-motivated, and determined to "beat their own time." The activity encouraged discipline, persistence, and a sense of healthy competition — qualities often difficult to cultivate through traditional instruction alone.

These findings confirm that *Count Me In TT* is more than a math game. It is a practical, classroom-ready tool that supports fluency development, focus, and confidence, while fostering a positive emotional connection to learning. With continued use, it has strong potential for integration into regular math programs across schools in Trinidad and Tobago.

COUNT ME IN TT - STUDENT RECORDS RAW DATA 2021 - 2025

✅ Key Takeaways

- **Every single student improved.**
- Average improvement: **~50% faster times**.
- **STUDENT 2 (71.9%)** and **STUDENT 7 (69.7%)** showed the biggest gains.
- Even students starting slower (20 minutes) finished near **6 minutes** after consistent attempts.

Student	Attempts	First Attempt	Last Attempt	Time Reduced	% Improvement
STUDENT 1	6	11:21.87	04:50.40	06:31.47	57.4%
STUDENT 2	8	20:13.25	05:40.94	14:32.31	71.9%
STUDENT 3	4	13:00.60	07:57.87	05:02.73	38.7%
STUDENT 4	7	09:42.53	06:36.81	03:05.72	31.9%
STUDENT 5	5	12:59.81	06:44.94	06:14.87	48.0%
STUDENT 6	6	09:14.48	06:06.48	03:08.00	34.0%
STUDENT 7	14	20:13.46	06:07.79	14:05.67	69.7%
STUDENT 8	15	14:38.32	05:50.62	08:47.70	60.0%
STUDENT 9	6	11:15.93	06:52.71	04:23.22	39.0%

Table showing a summary of 9 students' records of using Count Me In TT - Test Phase

OTHER FINDINGS

During four years of testing Count Me In TT with primary school students, the results went far beyond faster times. Students showed a visible change in attitude toward learning — growing more confident, motivated, and engaged with mathematics.

As they repeated the exercise, they began competing eagerly against their own previous times, celebrating each personal best. What started as a practice task quickly became a friendly challenge they looked forward to.

The game fostered enthusiasm, enjoyment, and resilience, transforming how students approached math. Many who once expressed frustration or boredom began saying things like “Sir, can we do the table today?” — a clear sign that learning had become fun.

INDIVIDUAL STUDENT RECORDS

(Student's full name not included for privacy reasons. All students are/were clients of A's Online Tutoring Services.)

PRIYA - STUDENT 1 (std 4 - 5)

DATE	TIME COMPLETED
25 June 2024	11:21.87
31 July 2024	16:37.61
27 August 2024	10:25.35
14 March 2025	6:40.15
30 Aug 2025	6:03.32
22 Sep 2025	04:50.40

AVA - STUDENT 2 (std 4 - 5)

DATE	TIME COMPLETED
09 Nov 2024	20:13.25
14 Dec 2024	13:10.70
04 Jun 2025	08:27.78
02 Jul 2025	08:06.11
30 Jul 2025	7:18.36
25 Aug 2025	7:08:45
3 Sep 2025	6:14.91
24 Sep 2025	05:40.94

MAYA - STUDENT 3 (std 4 - 5)

DATE	TIME COMPLETED
22 June 2024	13:00.60
6th August 2024	11:39.34
9th November 2024	08:54.14
23 January 2025	07:57.87

MAIAH - STUDENT 4 (std 3 - 4)

DATE	TIME COMPLETED
23 Sep 2023	9:42.53
14 Oct 2023	7:36.56
11 Nov 2023	7:31.75
9 Dec 2023	7:29.05
6 Jan 2024	6:55.14
3 Feb 2024	6:07.20
2 Mar 2024	6:36.81

GABRIEL - STUDENT 5 (Std 5)

DATE	TIME COMPLETED
12 Jan 2022	12:59.81
9 Feb 2022	11:16.16
9 Mar 2022	10:02.04
3 Jun 2022	10:16.34
1 Jul 2022	6:44.94

ALASJA - STUDENT 6 (Std 4 - 5)

DATE	TIME COMPLETED
8 JUN 2022	9:14.48
5 JUL 2022	7:06.19
8 SEP 2022	8:21.72
6 OCT 2022	6:05.82
10 NOV 2022	5:35.97
11 MAR 2023	6:06.48

JEN - STUDENT 7 (std 3 - 5)

DATE	TIME COMPLETED
30 OCT 2021	20:13.46
29 Nov 2021	14:41.42
16 Dec 2021	14:04.17
12 Jan 2022	17:13.90
2 Feb 2022	11:24.71
9 Mar 2022	13:15.92
7 Apr 2022	9:54.63
6 Oct 2022	13:16.19
19 Nov 2022	8:33.80
11 Feb 2023	8:00.51
11 Mar 2023	6:34.60
29 Apr 2023	6:25.84
10 Jun 2023	8:09.75
3 Feb 2024	6:07.79

JANAE - STUDENT 8 (std 3 - 5)

DATE	TIME COMPLETED
18 July 2022	14:38.32
18 Aug 2022	15.12.93
16 Feb 2023	12:19.13
6 Mar 2023	9:03.92
27 Apr 2023	9:45.50
23 May 2023	9:08.75
13 Jun 2023	7:00.0
11 Oct 2023	8:38.31
6 Nov 2023	6:36.09
10 Jan 2024	6:02.51
6 Feb 2024	6:18.05
5 Mar 2024	5:09.38
7 May 2024	6:25.47
1 Jun 2024	5:16.17
17 Jul 2024	5:12.73
1 Apr 2025	5:50.62

SHAKIAH - STUDENT 9 (std 4 - 5)

DATE	TIME COMPLETED
18 Jul 2022	11:15.93
16 Aug 2022	8:15.43
10 Sep 2022	7:46.34
15 Oct 2022	7:01.54
15 Dec 2022	7:48.16
11 Feb 2023	6:52.71

Recommended Completion Times by Level

Level	Typical Age	Target Time Range (minutes)	Goal Description
Standard 1	6–7 yrs	🥇 **12 - 15** 🥈 **15 - 20** 🥉 **20 - 30**	Still developing basic multiplication recall; focus on accuracy first.
Standard 2	7–8 yrs	🥇 **9 - 12** 🥈 **12 - 15** 🥉 **15 - 25**	Starting to memorize tables 1–12; can maintain steady rhythm.
Standard 3	8–9 yrs	🥇 **7 - 9** 🥈 **9 - 12** 🥉 **12 - 18**	Most tables memorized; aim to increase speed confidently.
Standard 4	9–10 yrs	🥇 **6 - 8** 🥈 **8 - 10** 🥉 **10 - 12**	Fluency improving; consistent practice brings sharp timing.
Standard 5 (SEA level)	10–11 yrs	🥇 **5 - 6** 🥈 **6 - 7** 🥉 **7 - 10**	Full mastery expected; focus on accuracy under timed pressure.
Form 1 (Secondary Transition)	11–12 yrs	🥇 **2 - 3** 🥈 **3 - 5** 🥉 **5 - 8**	Should demonstrate automatic recall and focus; ideal baseline for advanced math work.

Table showing recommended completion times for students.

Identifying Strengths and Weaknesses

An important secondary benefit of *Count Me In TT* emerged during the testing phase — its use as a diagnostic tool.
By observing students' live gameplay, both tutors and students were able to identify which multiplication facts caused hesitation or delay.
The points at which students paused, hesitated, or rechecked their answers revealed exactly where additional practice was needed.

This real-time feedback allowed learners to become more self-aware of their weaknesses and track their progress as they strengthened those specific areas.
It provided an efficient and engaging way to monitor fluency development without the need for formal testing.

Building Recall Through Practice

Students are encouraged to **look at the answers while they play**—especially in the early stages. This helps them recognize **patterns and relationships** between numbers, reinforcing understanding rather than rote memorization.

By visually connecting multiplication facts (for example, noticing that 6×7 and 7×6 give the same result), students develop **strong recall, pattern awareness, and number sense**.
Once these facts are memorized, they naturally begin to **focus on speed and fluency**, transitioning from learning to mastery.

Recommendation for Frequency and Tracking

To see real improvement, students should **use Count Me In TT at least once a week** or **a minimum of once per month**.
The key is **consistency** — steady repetition trains both memory and focus.

Encourage students to **log their time after every attempt** and compare their progress. Tracking results builds self-awareness and helps them see that small, steady improvements lead to major gains over time.

Even on days when students don't "feel like it," remind them to play anyway. **Pushing through those low-motivation moments strengthens discipline and perseverance**, qualities that extend far beyond mathematics.

Allowing each student to **set a personal target time** gives them ownership of their learning. When they set their own goal — and then beat it — they experience a real sense of achievement that builds confidence and intrinsic motivation.

Together, these habits — **consistency, tracking, resilience, and goal-setting** — form the foundation of lasting math fluency and self-belief.

Conclusion

The findings from this 2021–2025 testing phase confirm that *Count Me In TT* is more than a multiplication practice game — it is a **powerful learning tool** that enhances fluency, confidence, and motivation among students. Through consistent use, learners demonstrated measurable improvement in speed, accuracy, and self-awareness of their strengths and weaknesses.

Noticeable changes in attitude was observed— students began to approach mathematics with enthusiasm, determination, and a genuine sense of accomplishment as they competed to improve their times. This gamified approach to repetition transformed practice into play, making mathematics both meaningful and enjoyable.

The data also highlights *Count Me In TT*'s potential for **nationwide classroom integration**, offering teachers a free, accessible, and data-driven method to strengthen numeracy skills. As students engage regularly, the tool not only improves their performance but also cultivates discipline, goal-setting, and intrinsic motivation — key ingredients for lifelong learning success.

Appendix / Teacher Resources

To help teachers integrate Count Me In TT into classroom learning, the following resources are provided:

- Printable Leaderboard Template – For displaying student progress and encouraging friendly competition in class.
- Blank Progress Tracker Sheet – Allows students to record their times, set personal goals, and reflect on their improvement over time.
- Quick Access – **Scan the QR code** or visit countmeintt.com/leaderboard to view national standings and inspire participation.

Data Privacy & Parental Consent – All student submissions are voluntary and require parental or teacher permission. No personal data beyond name, and time are mandatory fields and are collected and stored. Other information such as school, and email are optional fields. This additional information is used as forms of contact for players who submit their times who win challenge prizes. Only Name, School (if entered) and time, is shared publicly on the Count Me In TT Leaderboard.